



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI Codebase Doctor: An AI-Powered Developer Tool for Automated Detection, Explanation, Repair and Verified Improvement of Software Repositories	
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Abstract

Modern digital environments generate enormous amounts of information across documents, databases, applications, social platforms and other data sources, making it increasingly difficult for users to locate specific information quickly and accurately. Conventional search systems primarily depend on keyword matching and predefined filters, which can miss relevant information, fail to understand the context of a query, and require users to manually examine large amounts of data. The proposed project, TraceFindAI, is an AI-powered intelligent information discovery and tracing system designed to overcome these limitations by combining automated data processing, semantic search and Large Language Model-based reasoning. Implemented as a Python-based application with a FastAPI backend, the system collects and processes information from connected data sources, converts unstructured content into searchable representations, and uses Natural Language Processing techniques to understand the semantic meaning of user queries. Relevant information is retrieved using vector embeddings and similarity-based search rather than relying only on exact keyword matches. The AI model then analyzes the retrieved context to identify related records, summarize important information, establish connections between different pieces of data and present the results in a clear and understandable form. TraceFindAI can also maintain a trace of the information sources used for generating each result, improving transparency and allowing users to verify the retrieved information. The system is designed to reduce manual searching effort, improve information retrieval accuracy and support faster decision-making in large and complex datasets. By integrating artificial intelligence, semantic search, Natural Language Processing and automated information tracing into a single platform, TraceFindAI provides an intelligent alternative to conventional search systems and enables users to discover relevant information through natural-language queries while maintaining context, traceability and explainability.

Signature of the Guide

HOD